

Comprehensive Briefing on IP Fragmentation and Problem Solving

Summary

IP fragmentation ensures that large IP datagrams can traverse links with smaller **Maximum Transmission Units (MTUs)**. When:

Datagram Size > MTU
and DF = 0

the router divides the datagram into fragments.

Core technical principles:

- IP Datagram = IP Header + Data
- Fragment Offset uses **8-byte scaling**
- All fragments except the last must carry data divisible by 8
- MF and DF bits control sequencing and permission
- Simple “divide and ceil” shortcuts may fail in edge cases

1. Fundamental Definitions and Components

1.1 IP Datagram Structure

At the Network Layer:

IP Datagram = Header + Data

Unless explicitly stated otherwise, any size mentioned for a datagram includes the IP header.

Important:

If a Transport Layer segment is given, that entire segment (Transport header + Transport data) becomes the “Data” field of the IP datagram.

1.2 Maximum Transmission Unit (MTU)

MTU defines:

Maximum size of (Header + Data) allowed on a link.

If:

Datagram Size $\leq$ MTU $\rightarrow$ Forward normally

Datagram Size $>$ MTU $\rightarrow$ Fragment (if DF = 0)

1.3 Fragmentation Control Fields

Header Length (HLEN)

- Stored in units of 4 bytes
- Header Size = HLEN $\times$ 4

Examples:

HLEN = 5 $\rightarrow$ 20 bytes

HLEN = 10 $\rightarrow$ 40 bytes

Total Length

Represents:

Header + Data of current fragment

Fragment Offset

Indicates:

Number of data bytes preceding this fragment

Stored as:

Offset = (Data Bytes Ahead) / 8

More Fragments (MF)

- $MF = 1 \rightarrow$ More fragments follow
- $MF = 0 \rightarrow$ Last fragment

Don't Fragment (DF)

- $DF = 1 \rightarrow$ Fragmentation prohibited
- $DF = 0 \rightarrow$ Fragmentation allowed

2. Theoretical Principles of Fragmentation

2.1 The 8-Byte Divisibility Rule

Because:

Fragment Offset is 13 bits

Total Length is 16 bits

Offset is scaled in units of 8 bytes.

Therefore:

Every fragment except the last must have:

$\text{Data Size} \% 8 = 0$

If not divisible:

Round down to nearest multiple of 8.

2.2 Reassembly Location

Reassembly occurs only at destination.

Reasons:

- Fragments may take different routes
- Router reassembly is computationally expensive
- A fragment may require further fragmentation downstream

Routers never merge fragments.

2.3 Fragment Identification Logic

Type	Offset	MF
First Fragment	0	1
Middle Fragment	> 0	1
Last Fragment	> 0	0
Unfragmented	0	0

3. Advanced Problem-Solving Insights

3.1 Transport vs Network Layer Data

If given:

Transport Layer PDU = 2260 bytes

UDP header = 8 bytes

IP header = 20 bytes

Then:

Network layer sees 2268 bytes of data (2260 + 8).

Total IP datagram size:

$2268 + 20 = 2288$ bytes.

Always clarify which layer the size refers to.

3.2 The “Shortcut” Fallacy

Common shortcut:

Number of fragments = $\text{ceil}(\text{Total Data} / \text{Max Data per Fragment})$

But this may fail.

Why?

Because:

Last fragment does NOT need to follow 8-byte divisibility rule.

Example Insight

MTU = 400

Header = 20

Available data space = 380

380 is not divisible by 8.

Typical adjustment:

Use 376 bytes per fragment.

But:

If remaining data ≤ 380 , even if not divisible by 8, it can fit entirely in final fragment.

Thus shortcut may overestimate fragment count.

Correct approach:

Step-by-step packing is safer.

3.3 Calculating Byte Ranges in Fragment

Given Offset:

First Byte Number = $\text{Offset} \times 8$

Last Byte Number =
 $(\text{Offset} \times 8) + \text{Data Length} - 1$

Example:

Offset = 100

First byte = 800

If data = 200 bytes:

Last byte = $800 + 200 - 1 = 999$

4. Structured Calculation Method

When solving fragmentation problems:

Step 1: Identify Given Quantities

- Total Data
- Header Size
- MTU
- Whether size refers to segment or datagram

Step 2: Compute Maximum Data per Fragment

Max Data = MTU – Header

If not divisible by 8:

Adjusted Max Data = $\text{floor}(\text{Max Data} / 8) \times 8$

Step 3: Generate Fragments Sequentially

Subtract data fragment-by-fragment.

For each fragment:

- Assign Offset = Previous Data Sent / 8

- Set MF = 1 (except last)

Step 4: Handle Last Fragment

Last fragment:

- Takes remaining data
- May violate 8-byte rule
- MF = 0

5. Instructional Pattern Analysis

Pattern 1: Small MTU Case

1000-byte datagram

20-byte header

MTU = 100

Available data = 80 bytes

80 divisible by 8 → valid

Fragments = 13

Pattern 2: Offset Interpretation

Offset = 300

Data ahead = $300 \times 8 = 2400$ bytes

If MF = 0 → last fragment

Pattern 3: Transport Encapsulation

8880-byte application data

UDP header = 8

IP header = 20

MTU = 1500

Total network data = 8888

Max per fragment:

$1500 - 20 = 1480$

Fragments = 7

Pattern 4: IP Overhead Calculation

If 6 fragments created:

IP Overhead = $6 \times 20 = 120$ bytes

6. Efficiency and Overhead Concepts

IP Overhead

Total header bytes across fragments:

Number of fragments $\times$ Header size

Fragmentation Overhead

Additional header cost beyond original:

$(n - 1) \times \text{Header Size}$

Efficiency Formula

Efficiency =
Original Data

7. Common Pitfalls

1. Ignoring 8-Byte Rule

Assigning offset without checking divisibility.

2. Forgetting Header Subtraction

Using MTU directly instead of $\text{MTU} - \text{Header}$.

3. HLEN Mistakes

Not multiplying HLEN by 4.

4. Terminology Confusion

Payload $\neq$ Total Packet
Segment $\neq$ Datagram

5. Miscounting Fragments

Blind ceiling division without considering last-fragment exception.